Performance Analysis of Deep Learning Models in Organic and Recyclable Waste Classification

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***Abstract*— Such technological advancements at such a wide level drives one counsel away in an unsustainable displacement effect. Waste disposal officials toil each day with inexpensive throwaway goods, like baby diapers, even as more green areas are being created in important cities. The toughest nut to crack is leftovers that cannot be composted or recycled. To solve this problem, a general automated deep learning waste classification system is required to improve the efficiency and accuracy of recycling systems UN. For better model performance, data from various sources were preprocessed via background removal, resizing and augmentation. In doing so, we need to have a good split for our dataset where we had 78% training, 20% testing and 2% validation data. Selected five models CNN, MobileNetV4, YOLOv8 EfficientNetB0 and ResNet50. A confusion matrix based on accuracy, precision, recall and F1-score was used to evaluate performance. The tests show the best performance from MobileNetV4 with an accuracy of ~98% across evaluation metrics. The next best are various versions of the YOLOv8, where we achieved 97–98% accuracy while taking advantages of running it in real time detecting object. Finally, this study concludes that MobileNetV4 and YOLOv8 are the best candidates for creating a simple, rapid, accurate, intelligent and improved classification system.**

Keyword — Waste Classification, Deep Learning, MobileNetV4, YOLOv8, Waste Management, Image Classification, Recycling System

I. Introduction

Municipal solid waste management is a longstanding challenge for sustainable development, especially in densely populated areas. This generation of waste has increased over time. Moreover, ineffective management practices could result in environmental contamination and health hazards for surrounding communities [1]. One key contributing factor is the ignorance of citizens in sorting out waste [3].

When organic waste and recycling materials are improperly separated, they get mixed together; and this makes the entire recycling process a lot more complicated. As a result, materials that could have been recycled avoid landfills [4]. Conventional waste sorting techniques also have serious challenges; effectiveness largely depends on the knowledge and work consistency of laborers, while biohazardous material poses a health risk due to direct contact with workers [5]. Due to these limitations, a fully automated system for classification and segregation of waste, in an efficient manner is required.

Recently, deep learning methods have been develop quickly especially in the context of data processing. These models can automatically process important visual characteristics and use them for classification. Histopathological images using deep learning techniques under the following categories based on its architectures Convolutional Neural Networks (CNN), MobileNetV4 and YOLO [7]. MobileNetV4 is known for being compact and accurate, which makes it ideal for use on devices with limited computational power. On the other hand, YOLOv8 is one of the most widely used since it detects and classifies objects fast and precise [9]. Since there are differences in design and the performance of these architectures, a comparative analysis is required to identify the best model with respect to classification accuracy and computational efficiency [10].

Hence this work uses five deep learning architectures such as CNN, MobileNetV4, YOLOv8, EfficientNetB0, and ResNet50 for the classification of organic waste verses recyclable waste. The goal is to find the model which provides the best trade-off between classification performance and efficiency. The rest of this paper is organized as follows: the section II reviews related works on deep learning for waste classification. Section III describes the methodologies of research, DATASET, data preprocessing and training and performance metrics for each model. Section IV provides the test results and a discussion comparing the models. Finally, we finish the work in Section V with final conclusions.

II. Literature Review

Some earlier research has corroborated the efficiency of deep learning for waste classification. For instance, Sharma et al. In Ref. [11], Wu et al. used CNN models to classify multiple types of wastes by replacing the traditional algorithms through learning more high-resolution visual features from data. Nguyen et al. As pointed out by [12], EfficientNetB0 gives better accuracy with less number of parameters as compared to standard CNN models. Similarly,

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Rahman et al. MobileNetV4 [13] exhibits a decent state of the art in both classification and computational efficiency (that makes it suitable for mobile applications). Kumar et al. [14] Working with the YOLO model they implemented a Real Time Waste Detection system and got quite good results. Meanwhile, Li et al. Finally, [15] provides an comparison between ResNet50 and EfficientNetB0, it shows EfficientNetB0 perform good efficiency while have high accuracy. While these studies show promising results, they evaluated only one or two model types each, precluding clear comparisons; hence, a comprehensive evaluation over more deep learning models is warranted.

TABLE I. RESEARCH GAP COMPARISON

Study	CNN	ResNet50	EffNetB0	MobileNetV4	YoloV8
Schema [11]	✓	×	×	×	×
Nguyen [12]	×	×	✓	×	×
Rahman [13]	×	×	×	✓	×
Kumar [14]	×	×	×	×	✓
Li [15]	×	✓	✓	×	×
Proposed	✓	✓	✓	✓	✓

No previous research has been performed in the same waste classification dataset at the same time of CNN, ResNet50, EfficientNetB0, MobileNetV4 and YOLOv8 (Table I). Existing studies usually develop only a few models, and the experimental results vary considerably. To overcome this issue, the current study compares all five models using same dataset, data preprocessing and evaluation techniques. Apart from classification results, model complexity is evaluated in terms of parameters, training time, inference time and FLOPs.

III. RESEARCH METHOD

The study consists of six phases. We first collect and preprocess the waste data before we use this data for model training. Then, this paper trains five deep learning architectures CNN, MobileNetV4, YOLOv8, EfficientNetB0 and ResNet50 to classify images into organic and recyclable waste. Various metrics such as accuracy, precision, recall and F1-score are used to compare the effectiveness of each model in order to choose the best architecture for some datasets.

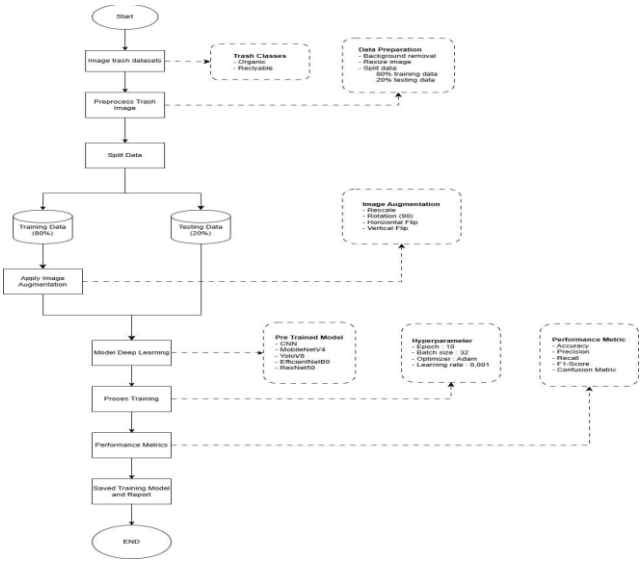


Fig. 1. Research diagrams for the study of trash image classification uisng several stages

In Figure 1, we illustrate a flowchart of image-based waste classification research using stages that include starting with collecting dataset images followed by preprocessing steps that help in the enhancement of quality characteristics through mechanisms such as background removal and image resizing. Then you split the data into 78% used for training, 2% validation and 20% to test it, where the ANN which uses the augmenting process on the training data in order to have more variation of the same data. This data is used in the deep learning modeling session with multiple models (CNN, MobileNetV4 YOLOv8 EfficientNetB0 ResNet50). Models are trained by specifying different parameters followed by evaluation based on metrics like accuracy, precision, recall, f1-score and confusion matrix which in turn gives a trained model along with the report as an output of research.

A. Data Collection

The data used in this work came from different Waste Classification datasets found on Kaggle, Roboflow Universe and other public sources. Images from various data sources underwent manual inspection and subsequently re-labelling into two classes (recyclable waste, organic waste), followed by consolidation of these images into a single dataset wardens were excluded to improve quality duplicate and low-quality images were removed. Here is a description of the dataset categories applied throughout this study.

TABLE II. CATEGORY DATASETS

Category	Object Example	Characteristic
Organic	Food scarps, leaves, and fruit peels	Easily decomposed (Biodegradable)
Recyclable	Plastic bottles, paper, cans, glass	Recyclable

To perform classification, Table II was divided into two classes from the dataset to make it easier and follow technology based waste management systems adopted in the literature. Additionally, we focus on this classification since it

is faster and computationally more efficient than multi-class classification approaches



Fig. 2 Waste dataset image from data classification

Figure 2 displays the waste image dataset, that includes a total of 21,076 images established into three subsets (Train, Validation and Test). The training set: consists of 16,439 images (78%) and used to learn the parameters of the model. The validation set containing 422 images (2%) is used in the training process to monitor model performance, fine-tune hyperparameters and prevent overfitting. The evaluation set consists of 4,215 images (20%), which are used only for the final evaluation step of our trained models. To evaluate the classification performance of several deep learning models, this train-test dataset split has been applied consistently to all CNN, MobileNetV4, YOLOv8 EfficientNetB0 and ResNet50 architectures. The dataset used in this study was pulled from an open-access data repository on Kaggle. It deals with two major types of waste, organic waste and recyclable waste. The dataset consists of 21,076 images, where organic has 10,262 images and recyclable waste has 10,814 images. For consistency, all samples were collected from a single source for distribution and labeling. Because the data had already been grouped into two target groups, it did not require any information from other datasets. This setup provides data uniformity which allows the training and relative performance comparison of all models tested on a like for like basis.

B. *Dataset Processing*

The collection of a waste image dataset from public sources started the data preparation process. Then, the background from each image was removed as a preprocessing step to eliminate extraneous noise and keep viewers focused on the top object of interest. Then, all images had to be resized based on the input size fixed in each deep learning model. After preprocessing, the data was split into a training 78%, testing 20% and validation set 2%. Using the training data, I created five models: CNN, MobileNetV4, YOLOv8, EfficientNetB0 and ResNet50. At this stage, each model learned to identify visual patterns that easily distinguished organic waste from recyclable waste by fine-tuning its internal settings.

The results in models were tested on accuracy, precision, recall and F – measure. Lastly, we selected the model that performed best over the previous phase for the final stage of implementation and analysis.

TABLE III. TRAINING HYPERPARAMETERS

Parameter	Value
Epoch	10
Batch Size	32
Optimize	Adam
Learning Rate	0.001
Loss Function	Binary Cross Entropy
Input Image Size	224×224

The training configurations for the different models to make sure a fair comparison is given are summarized in Table III. Batch size was set to 32 and we used Adam optimizer which does a decent job in keeping the learning process stable, allowing models to converge quickly. Binary cross-entropy was chosen for the loss function for this binary classification problem. Identical conditions were used to train all models for 10 epochs. On the other side, this relatively short training time may have limited deeper architectures such as ResNet50 and EfficientNetB0 to successfully converge properly.

All experiments are performed on Google Colab. An NVIDIA Tesla T4 GPU (16 GB) was supplied in the environment which included 12 Gb of system RAM with an Intel Xeon CPU. Software stack based on Python 3.10; tensor flow for the execution of CNN, MobilenetV4, EfficientNetB0, ResNet50 components and YOLOv8 implemented through the Ultralytics library. Fixing all hyperparameters across experiments allows us to evaluate both classification accuracy and computational efficiency fairly.

C. *Classification Using Deep Learning Algorithms*

This study is a comparative experimental design of five deep learning models acting on digital images to classify waste as organic and recyclable waste. The models used are simple-based architectures, lightweight models, deep networks based models as well as object-detection based-models.

In this study, five different models are utilized: Convolutional Neural Network (CNN), MobileNetV4, YOLOv8, EfficientNetB0 and ResNet50. These models were chosen to span a wide variety of deep learning architectural properties, so that we could learn about the performance of each model fully.

1. *Convolutional Neural Network (CNN)*

A Convolutional Neural network (CNN) is a type of deep learning neural network suitable for data classification. It will go through convolution and pooling stages to extract visual patterns, and then classification will be performed using fully connected layers. The CNN in this study is used as a baseline model for comparisons with other architectures.

$$f(x) = max(0, x) \quad (1)$$

where *f(x)* is the activation output and x is the input value [16].

2. *MobileNetV4*

MobileNetV4 is an optimized architecture for mobile devices and edge hardware with a very small size of the model. It uses depth wise separable convolution, a method

that reduces the computational burden but maintains classification accuracy.

$$Y_d(i,j,k) = \sum_{m=1}^{M}\sum_{n=1}^{N} X(i+m, j+n, k)K(m,n,k) \quad (2)$$

The resulting feature maps are then combined through pointwise convolution:

$$Y_p(i,j,l) = \sum_{k=1}^{C} Y_d(i,j,k)\ P(k,l) \quad (3)$$

This structure allows processing to be faster with fewer parameters than standard convolution [17].

3. YOLOv8

YOLOv8 is a one-stage detection model, which means it can locate and classify objects in a single step. Its architecture is very friendly to real-time, therefore this model can be used in an automated waste identification system. By the use of their definition of loss function over total model, the loss function in overall can be defined as!

$$Loss = L_{box} + L_{obj} + L_{cls} \quad (4)$$

where *Lbox* is relative to the localization loss while *Lboj* determines the objectless loss and *Lcls* refers to classification loss [18].

4. EfficientNetB0

EfficientNetB0 employs a balanced scaling strategy that leads to improved network performance. We can formalise such a method as follows

$$depth = \alpha^{\varphi}, width = \beta^{\varphi}, resolution = \gamma^{\varphi} \quad (5)$$

with constraints:

$$\propto \cdot \beta^{\varphi} \cdot \gamma^2 \approx 2 \quad (6)$$

This strategy is used by the model to obtain high-level accuracy with a relatively inexpensive number of parameters [19].

5. ResNet50

ResNet50 is a deep neural network model with the residual connections to counteract the vanishing gradient issue across shallower layers. You can express the function of residual as follows:

$$y = F(x, W) + x \quad (7)$$

where F(x, W) is the residual learning function and x is the input. This technique enables training of very deep networks without performance degradation [20].

D. Evaluation Matrix

In this research, the classification performance was measured using accuracy, precision, recall, and F1-score.

These metrics were derived from the confusion matrix, which consists of True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values [21].

Accuracy indicates the proportion of samples correctly classified out of the total data [22]:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (8)$$

Precision refers to the measurement of how many samples predicted as positive actually belong to the positive class [23]:

$$Precision = \frac{TP}{TP+FP} \quad (9)$$

Recall evaluates the model's ability to accurately detect positive samples [24]:

$$Recall = \frac{TP}{TP+FN} \quad (10)$$

F1-score combines precision and recall into a single metric, providing a balanced measure of classification performance [25]:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (11)$$

This enables comparison between the different deep learning models used for waste classification. The deep learning models used in this work were initialized with pre-trained weights on image datasets. Pre-trained models enable the networks to capitalize on visual representations they've already learned from large volumes of image data, which can speed up convergence and improve performance when applied to more modest datasets. For this study, transfer learning has been used to fine-tune the already existing networks rather than training them from scratch. The initial layers were kept frozen in order to preserve general feature extraction capabilities, and the terminal layers were retrained to allow for model adaptation for classifiying organic and recyclable waste. This fine-tuning strategy allows the models to learn dataset-specific features while keeping the robustness of the previously learned representations.

IV. Result and Discussion

This experiment evaluated five deep learning models for classifying waste images as organic and recyclable. The classification results were analyzed by accuracy, precision, recall, and F1-score metrics.

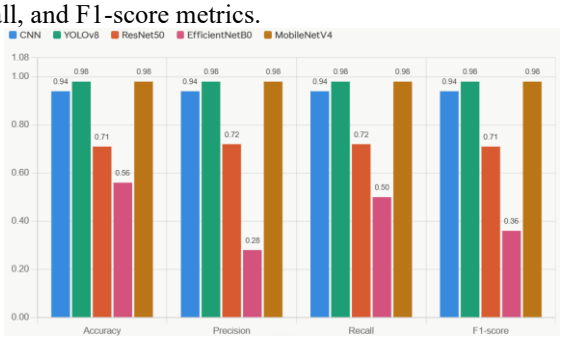


Fig 3. Precision comparison of CNN, YOLOv8, ResNet50, EfficientNetB0, and MobileNetV4

Bar chart in Figure 3 is convert these numbers in the visual comparison of four Evaluation Metrics. YOLOv8 and MobileNetV4 build two towering, almost identical bar charts, which visually verifies how consistently one performs over the other no matter which of the metrics we are considering. A very different story is told about EfficientNetB0 its precision and recall bars are positioned much lower than the other models, creating a w-formation that matches our zero scores coverage discussed above. The unequal pattern further corroborates the conclusion can be inferred table since, while multiple models achieved satisfactory performance for the two class waste classification task, not all of them do it in an equally balanced and reliable manner.

TABLE IV. MODEL COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
MobileNetV4	0.985	0.985	0.985	0.985
YOLOv8	0.954	0.964	0.954	0.959
CNN	0.941	0.941	0.943	0.942
ResNet50	0.724	0.724	0.723	0.723
EfficientNetB0	0.281	0.281	0.500	0.360

In Table IV, we show a comparison of the performance between the five deep learning models tested in our study. In particular, MobileNetV4 consistently achieved highest classification performance over various evaluation metrics, overall. YOLOv8 performed similarly well but has the advantage of also being real-time. A simple CNN gave satisfactory baseline performance, while ResNet50 and EfficientNetB0 struggled with fine-grained organic vs. recyclable waste categories.

TABLE V. COMPUTATIONAL COMPLEXITY AND EFFICIENCY COMPARISON

Model	Parameters (M)	FLOPs (G)	Training Time (10 Epoch)	Inference Time (ms/image)
MobileNetV4	3.8	0.40	6,851 s (114.2 min)	2.5–3.0
YOLOv8	2.7	0.8	7.495 h (449.7 min)	14.0
CNN	3.5	0.6	2,763 s (46.1 min)	3.85
ResNet50	25.6	4.1	2,705 s (45.1 min)	4.25
EfficientNetB0	5.3	0.39	21,309 s (355.2 min)	79.3

Table V summarizes the computational properties of the evaluated models. MobileNetV4 provided the least inference time requiring only 2.5–3.0 ms per image while achieving the best classification accuracy across all architectures that were tested in this study. Both CNN and ResNet50 achieved similar training times (45 to 46 minutes) while having inference speed below 5 ms/image. On the contrary, EfficientNetB0 achieved an average training time of 355 minutes at least and generated the slowest inference performance compared to classification models. YOLOv8 achieved a fair inference speed but the training took about 7.5 hours, however, this was running on a cpu based setup which contributed to the greater training time taken. These results suggest that on the combined tasks of organic and recyclable waste classification, MobileNetV4 is the most optimal when considering its classification correctness to compute cost ratio.

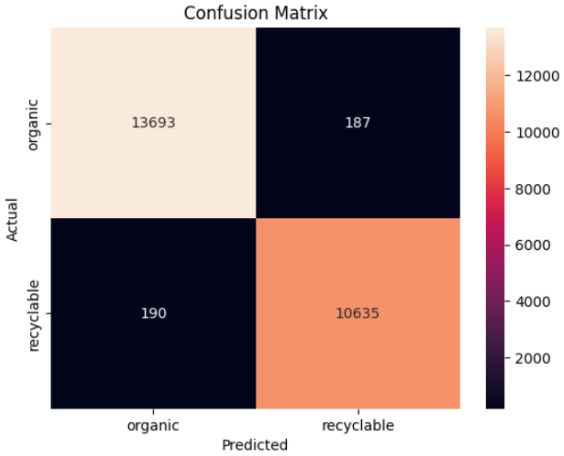


Fig 4. Confusion Matrix of MobileNetV4 Model on the Waste Classification Test Dataset.

The confusion matrix of MobileNetV4 model is shown in Figure 4. It means, these values were correctly predicted organic waste images of 13693 and recyclable waste images of 10635. In contrast, 187 organic images were falsely labelled as recyclable and 190 recyclable images were incorrectly classed as organic.

The low misclassified samples compared to the complete testing data showcases that MobileNetV4 manages a good separation of organic vs. recyclable waste. The confusion matrix above shows most predictions on the diagonal, indicating a very high level of classification accuracy. This result implies that MobileNetV4 lightweight architecture extract the relevant visual features along with efficient computational performance. There are no major biases towards one class or another near the performance of 0.5 and thus it can be stated that the errors evenly distributed amongst each class between those classes.

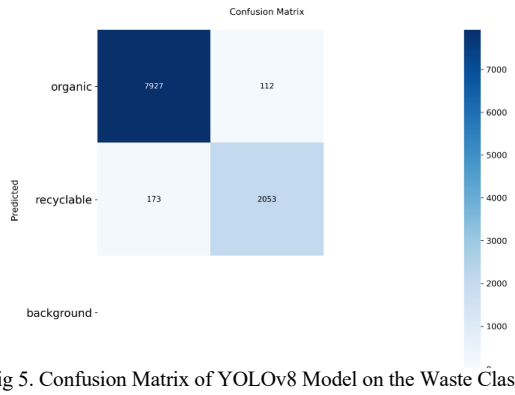


Fig 5. Confusion Matrix of YOLOv8 Model on the Waste Classification Test Dataset.

The confusion matrix of YOLOv8 model is shown in the Figure 5. The 7,927 organic waste images and 2,053 recyclable waste images were correctly classified by the model; however, 112 organic images were misclassified as recyclable dirt and 173 recyclable dirt images are misclassified as organic. The high incidence of correct predictions along the diagonal indicates that YOLOv8 performed well on classification. The largest proportion of classification mistakes were made on recyclable waste samples and could be due to confusion with visually similar

objects between recyclables and compost. Despite these errors, the overall number of misclassifications remained relatively low, indicating that YOLOv8 is capable of performing waste recognition effectively while maintaining the advantages of real-time detection.

While EfficientNetB0 produced no precision and recall for one class it seems that the model could not learn any features at all to identify this as a representative class. This resulted in the model predicting samples as one class fully dominating others, yielding poor classification performance. The problem could be attributed to improper hyperparameter choices, not training enough or simply that these classes of models have limited ability to express the characteristics of the dataset. The experiments indicate that the powerful EfficientNetB0 architecture is highly dependent on training configuration and dataset-specific optimization.

V. CONCLUSION

This study assessed the comparison of five deep learning models CNN, MobileNetV4, YOLOv8, ResNet50 and EfficientNetB0 in classifying waste images into organic and recyclable categories. Precision, recall and F1-score metrics were used to evaluate the performance of the models. Different models have been evaluated based on performance values, and MobileNetV4 has received the highest performance rating overall with results between 0.98–0.99 for both classes YOLOv8 obtained similar results and also takes the clear advantage of real-time object detection, enabling it to be implemented more realistically in waste-sorting tasks. The CNN performed reasonably well at classifying images and can be applied as a nice baseline model.

By contrast, ResNet50 yielded low evaluation scores illustrating it is not adapting well to the dataset. This indicates that EfficientNetB0 is not sufficient /overfitted and did very poorly in separating the waste categories. The overall results show that model architecture is an important factor for the accuracy of waste classification. Based on the experimental results, MobileNetV4 and YOLOv8 are the most promising approaches for developing reliable and efficient automated waste management systems.

AUTHOR CONTRIBUTION

This research was prepared and designed, including proposing the model and algorithm used by Hans Christian Kosasih. The writing of the paper was supervised by Shieren Revalina Jesslyn. The experiment and colleting the trash classification data was carried out by Virlene Marchella. Yasi Dani contributed as the supervisor for the research material and provided academic guidance throughout the study. All authors read and approved the manuscript.

DATA AVAILABILITY

The dataset used in this study is publicly available through Roboflow and can be accessed via the following link: *https://universe.roboflow.com/shierens-workspace/trash-data-classification*. The dataset consists of labeled images used for training and testing deep learning models to classify waste into organic and recyclable categories. All preprocessing steps, model training, and evaluation were conducted using Google Colab. The implementation code supporting the findings of this study is available from the corresponding author upon reasonable request to ensure reproducibility of the results

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